

Aditya Prakash

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Projects

Verdict | *TypeScript, React, Node.js, PostgreSQL, Prisma, BullMQ, WebSockets, Redis, Tailwind CSS.* **May 2026**

- Architected an asynchronous grading pipeline using Redis, BullMQ, and the Gemini API to process PDF submissions and rubric-based evaluations without blocking the main server thread.
- Engineered a decoupled real-time event streaming architecture using Redis Pub/Sub and WebSockets, broadcasting granular AI evaluation progress from isolated Python workers to connected clients.
- Designed presigned URL uploads with Cloudflare R2 (S3), offloading direct PDF transfers from the Node.js backend to reduce memory overhead and latency.
- Modeled a robust assignment evaluation lifecycle in PostgreSQL and Prisma, managing state transitions and attempt tracking to support re-evaluations and submission workflows.
- Repository Link: <https://github.com/adityasrc/verdict> | **Live**

Flowboard | *Next.js, TypeScript, Node.js, Express.js, PostgreSQL, Prisma, WebSockets, Tailwind CSS.* **Jan 2026 - Mar 2026**

- Engineered a collaborative whiteboard using Next.js, HTML5 Canvas, and **WebSockets**, decoupling real-time drawing from asynchronous database writes.
- Secured WebSocket authentication via Sec-WebSocket-Protocol headers and designed an in-memory room cache to reduce database lookups.
- Optimized real-time network utilization through distance-based coordinate filtering, reducing redundant drawing events while maintaining smooth collaboration.
- Designed state synchronization using UUID-based echo cancellation and user-scoped undo stacks for reliable concurrent collaboration.
- Repository Link: <https://github.com/adityasrc/flowboard> | **Live**

Training

Lovely Professional University – DSA **Jun 2025 - Jul 2025**

- Completed an intensive 6-week DSA program covering advanced data structures and graph algorithms.
- Developed a predictive model for electric grid failures using BFS/DFS graph traversals to identify critical failure nodes.
- View **Certificate**

Certificates

Bits and Bytes of Computer Networking **Nov 2024**

Coursera — Certificate Link

Introduction to Hardware and Operating Systems **Oct 2024**

Coursera — Certificate Link

Fundamentals of Network Communication **Sept 2024**

Coursera — Certificate Link

Skills

Languages: C++, Go, JavaScript, TypeScript

Frontend: React, Next.js, Tailwind CSS, HTML5 Canvas

Backend: Node.js, Express.js, REST APIs, WebSockets, Redis, BullMQ

Databases: PostgreSQL, MongoDB, MySQL

Tools: Git, GitHub, Postman, Docker, Turborepo

Education

Lovely Professional University **2023 – 2027**

Bachelor of Technology - Computer Science and Engineering — CGPA: 7.84

Phagwara, Punjab

St. John's Academy **2021 – 2022**

12th — Percentage: 75%

Vaishali, Bihar

St. Paul's High School **2019 – 2020**

10th — Percentage: 94.4%

Vaishali, Bihar